THE UNIVERSITY OF MELBOURNE

FNCE 30007 Derivative Securities

Lecture 1 – Introduction to Futures

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Outline

- Derivative definition
- Futures definition
- Settlement of futures contracts
- Relationship between futures price and spot price
- Margins and marking-to-market (daily settlement)
- Forward vs. Futures
- Regulation

Reading: Chapters 1 and 2

The nature of derivatives

- A derivative is an instrument whose price depends on, or is derived from, the price of another asset.

Examples of derivatives

- Agreements to buy or sell an asset some date in the future —*Futures and forwards*
- The right to choose to buy or sell an asset some date in the future — *Options*
- Exchanging period cash flows based on the realisation of some asset price — *Swaps*

Why study derivatives

- Purposes of using derivatives
  - Hedging
  - Speculation
  - Arbitrage for profits
- Markets for derivatives are huge
  - US$846 trillion in notional outstanding and US$21.8 trillion in gross market value for over-the-counter (OTC) derivatives as at June 2025 (see BIS)
- They can have disastrous consequences if misused
  - Most derivatives trades take place outside of centralised and regulated exchanges and clearing houses
  - Traders can take large amounts of *leverage* using derivatives
  - Example: the collapse of AIG based on collateral calls on written CDS contracts

The Australian Securities Exchange

1960: The **Sydney Futures Exchange (SFE)** was established to facilitate futures trading in Australia.

1990s: The **Australian Stock Exchange (ASX)** launched exchange-traded equity derivatives, including individual share options and share futures, alongside the SFE's derivatives market.

2006: The ASX and SFE merged to form the **Australian Securities Exchange (ASX)**, integrating Australia's cash equity and derivatives markets.

Today: The ASX offers a wide range of **exchange-traded derivatives**, including:

- Equity and index options
- Interest rate, equity index and commodity futures
- Warrants and exchange-traded options

Futures contracts

- A futures contract is an agreement to buy or sell an asset at a certain time in the future for a certain price
- By contrast, in a spot contract there is an agreement to buy or sell the asset immediately or within a very short period of time
- The futures price is the price at which you agree to buy or sell



Terminology

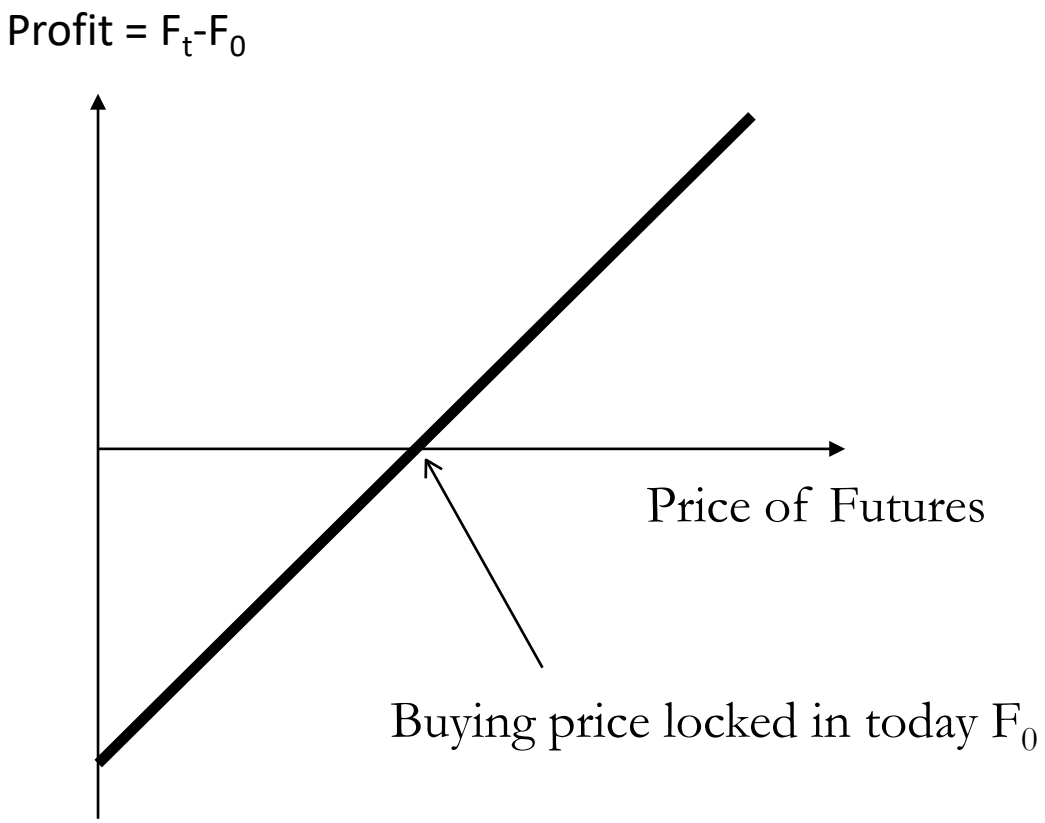
- A long futures position is an agreement to buy the asset at a certain time in the future for a certain price
- A short futures position is an agreement to sell the asset at a certain time in the future for a certain price

Examples of futures contracts

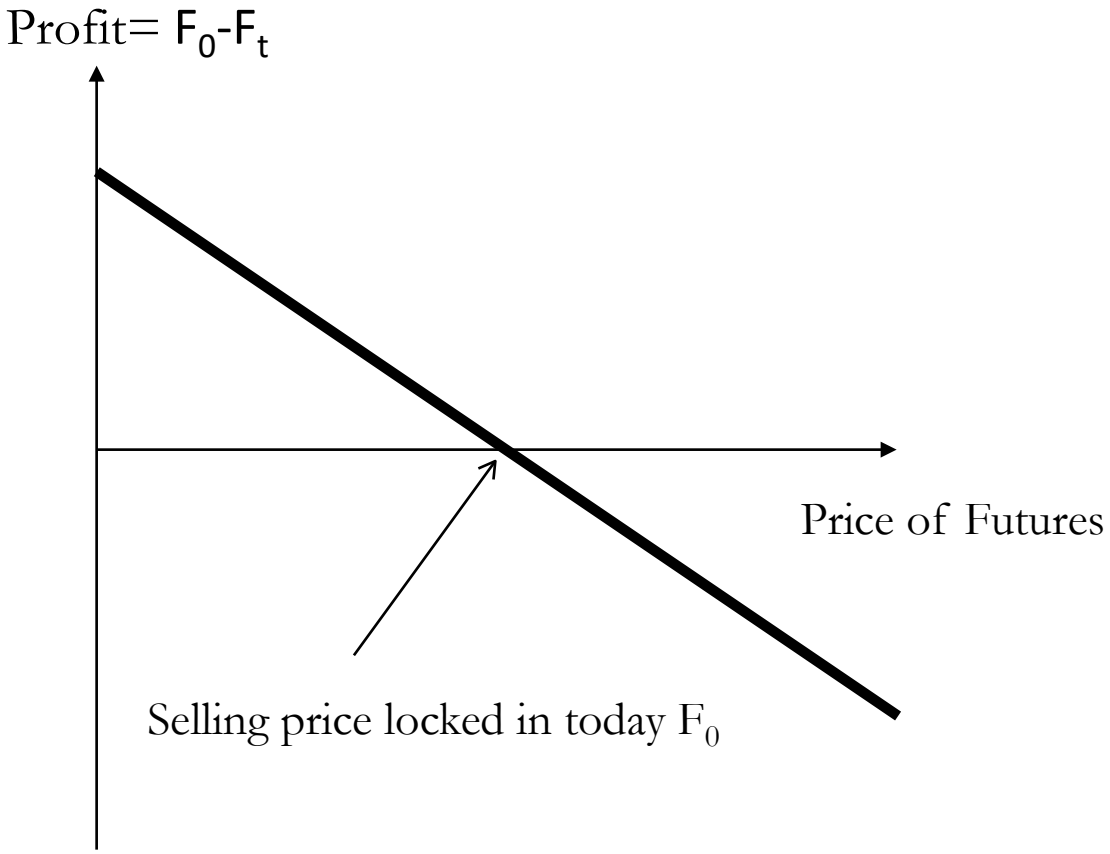
- Agreement to:
  - Buy 100 oz of gold @ US$1,050/oz in December
  - Sell £62,500 @ 1.5500 US$/£ in March
  - Sell 1,000 bbl of oil @ US$75/bbl in April
  - Buy 100 90-day bank bill futures contracts expiring in June

  - https://www.asx.com.au/markets/trade-our-derivatives-market/derivatives-market-prices/grain-derivatives

Profit from a Long Futures Position



Profit from a Short Futures Position



Settlement - Delivery

- A futures contract is an agreement to buy or sell an asset for a certain price at a certain time in the future
- It can be settled by delivering the assets underlying the contract. When there are alternatives as to what is delivered, where it is delivered, and when it is delivered, the party with the short position chooses

Cash settlement

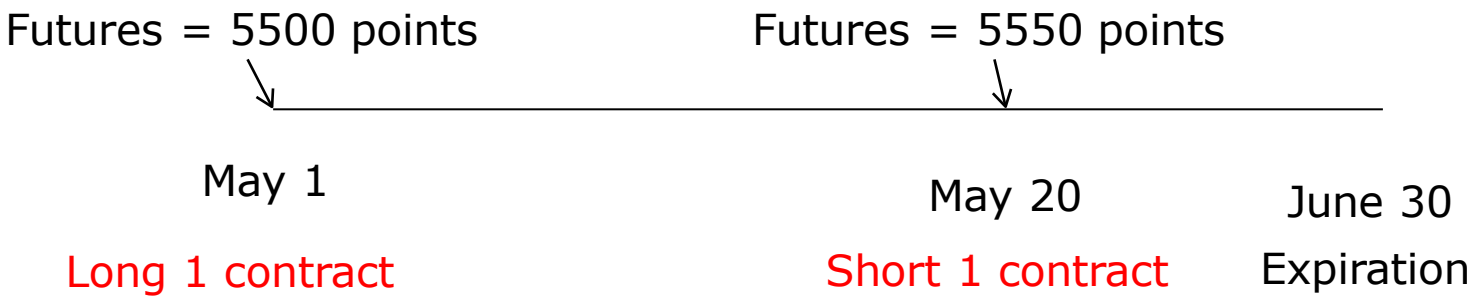
- Cash settlement happens often when it is inconvenient or impossible to deliver the underlying asset
- Final settlement price is equal to the spot price of the underlying asset on that day
- Price will still be locked in at the Futures price that both parties agreed on

Closing out futures positions

- Investors may close out their positions before settlement
- Closing out a position involves entering into an opposite trade to the original one that opened the position
  - For example, an investor who buys five July corn futures contracts on 6 May can close out the position on 20 June by selling five July corn futures contracts

Example: SPI 200 futures

- The SPI 200 futures contract tracks the price fluctuations of the ASX 200 index.
- Suppose an investor is bullish ⇨ long futures
- Each point move in the futures contract is worth $25
  - Long: contract price increases from 5500 to 5501 ⇨ $25 profit/contract
  - contract decreases from 5500 to 5499 ⇨ $25 loss/contract



Example: SPI 200 futures

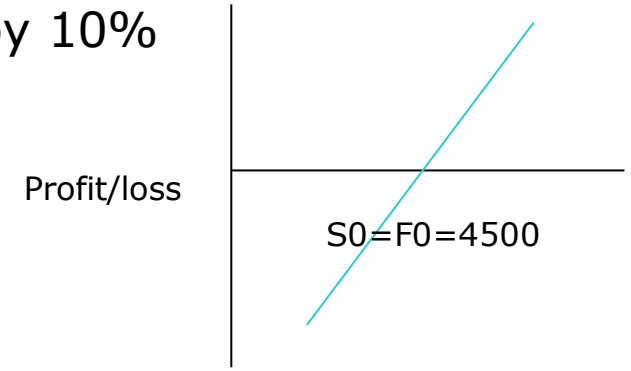
- Profit is 50 points x $25/point = $1250
- If the market falls to 5480 the investor loses 20 points x $25/point = $500.

- If the speculator was bearish they would short the index on May 1.
  - If index drops to 5480 ⇨ profit is $500/contract.
  - If index increases to 5550 ⇨ loss is $1250/contract.

- So why not speculate on the index through an ETF?

Example: SPI 200 futures

- Consider a bullish investor with $80K. The index and futures are currently at 4500 points ($S_0=F_0=4500$).
- Strategy 1 (Long Spot):
  - Invest $80K in an index fund (ETF)
- Strategy 2 (Long Futures):
  - Invest $80K in SPI futures.
  - Each contract requires $8K margin (more on this soon)
  - Go long 10 SPI futures contracts
  - Each point is $25
- Imagine the market increases by 10%



Example: SPI 200 futures

- Strategy 1: Spot position (ETF)
  - Returns of $80K*0.10 = $8K
  - Investment increases to $88K
- Strategy 2: Futures position
  - Each contract increases by 10% from 4500 to 4950.
  - Profit per contract = 450 points * $25 = $11,250
  - Total profit = 10 contracts * $11,250 = $112,500
  - Investment increases to $192,500

  - Another perspective
  - Value of exposure = 4500points*$25*10 contracts = $1,125,000
  - Increase of 10% = $112,500

Convergence of Futures Price to Spot Price

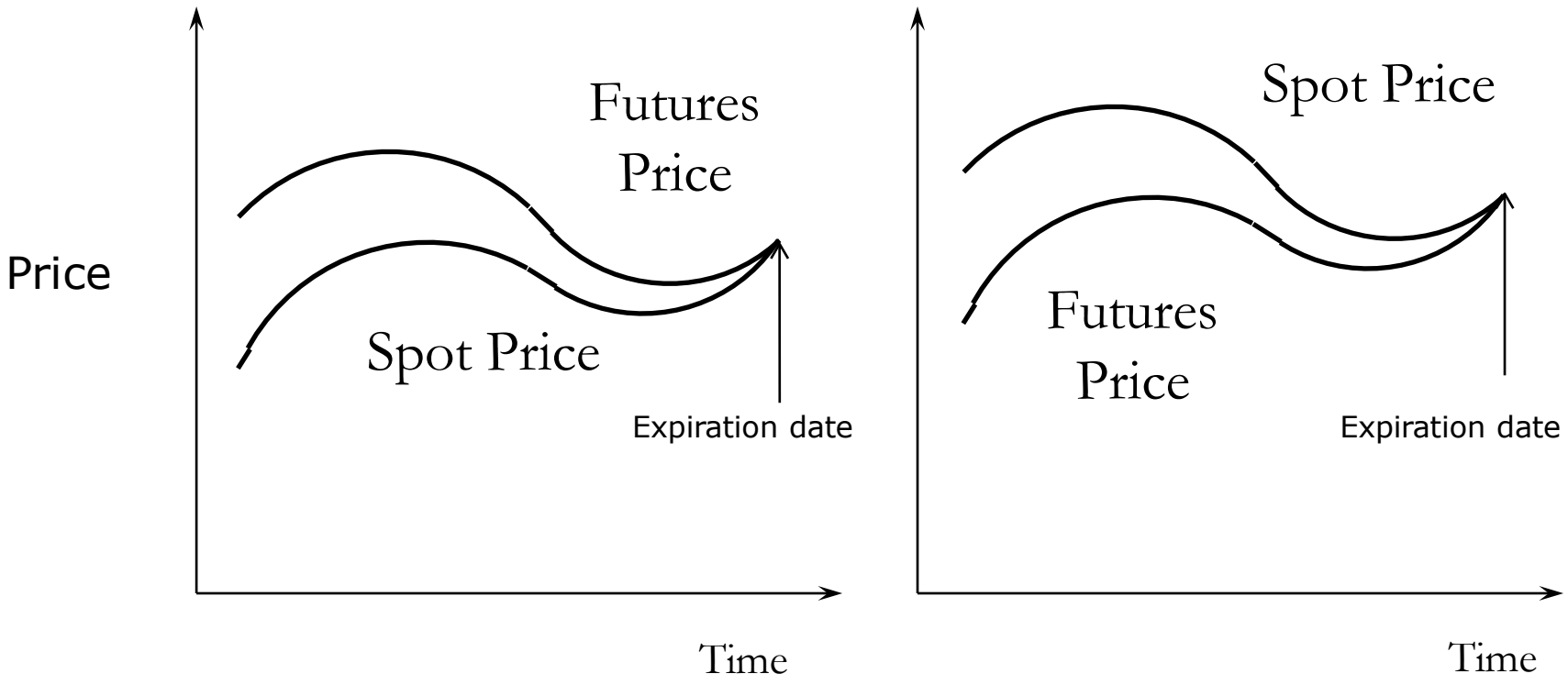
- As futures approaches expiration ⇨ futures price converges to the spot price.
- Otherwise there's an arbitrage opportunity.

Example:

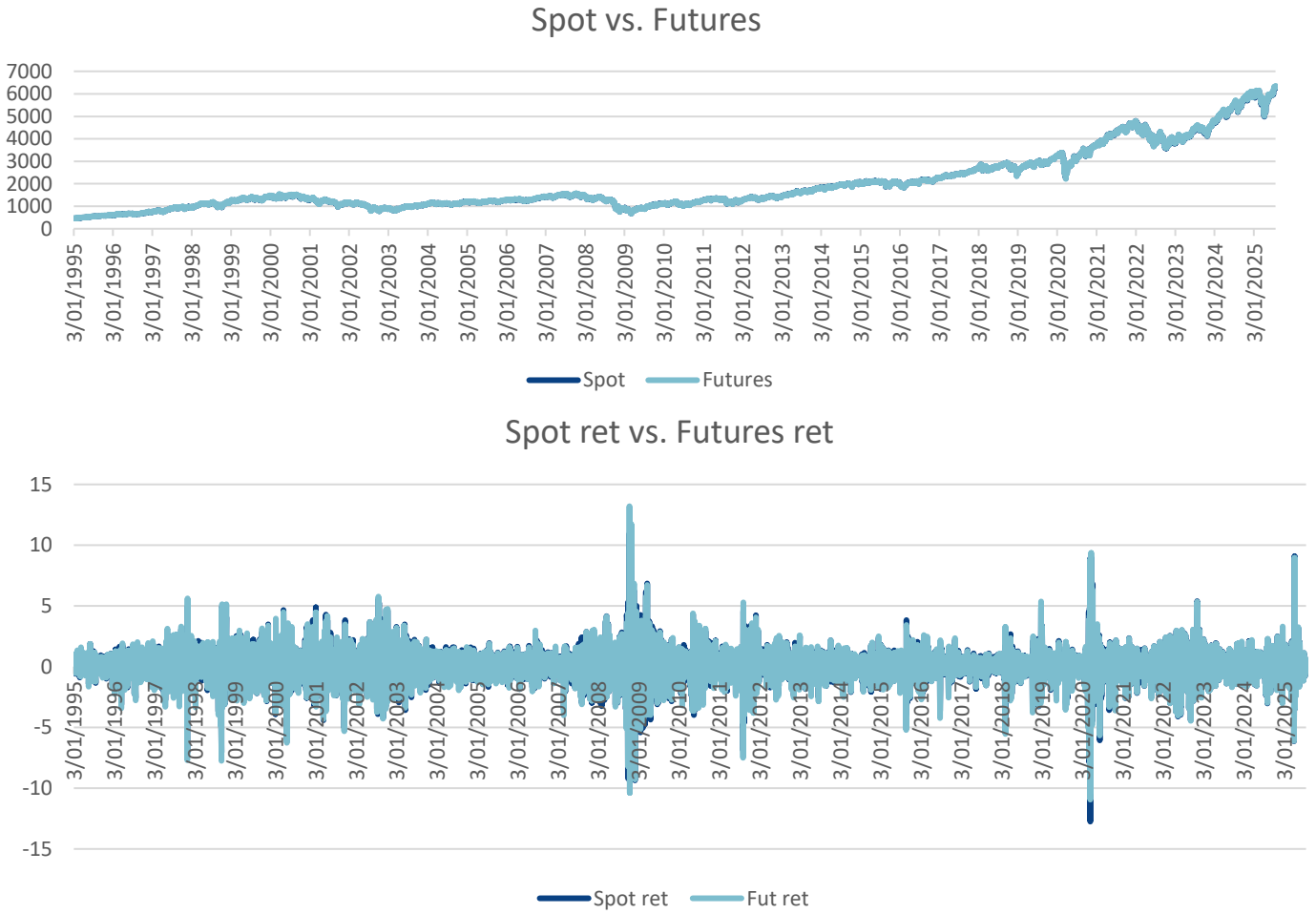
Assume futures is above the spot at maturity $F_T > S_T$

- sell overvalued security and buy undervalued one.
- Arbitrageurs sell (short) a futures contract, buy the asset, and make the delivery.
- ⇨ Futures price ⇩ & spot price ⇧
- Continue until prices are equal (subject to transaction costs)

Convergence of Futures Price to Spot Price

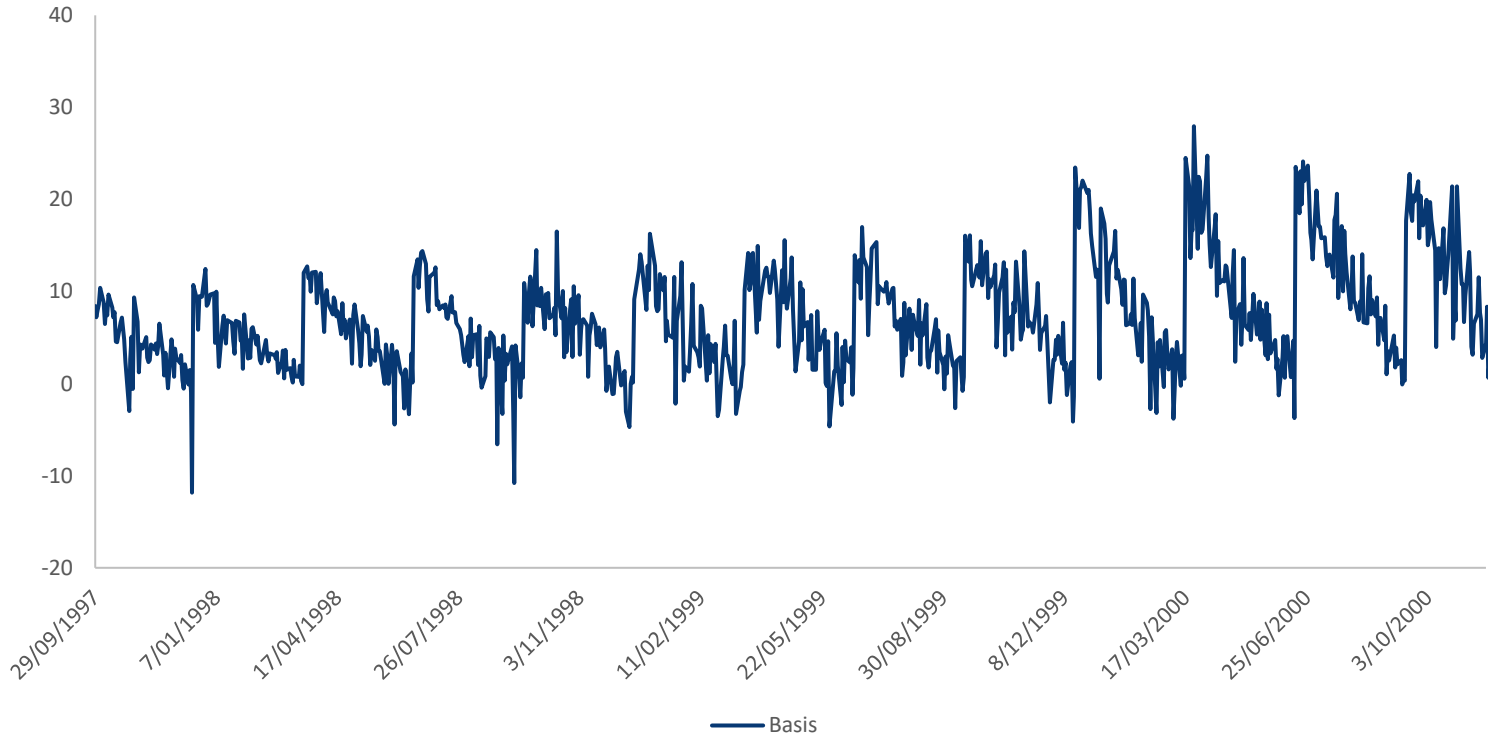


Convergence of Futures Price to Spot Price



S&P500 futures basis(t) on the nearby = F(t)-S(t)

Convergence of Futures Price to Spot Price



The clearinghouse and clearing margins

- A clearing house acts as an intermediary in futures transactions. It guarantees the performance of the parties to each transaction.
- The clearinghouse keeps track of all the transactions that take place so that it can calculate the daily net position of each of its members.
- Brokers are required to maintain margin accounts with clearinghouse members and brokers are required to maintain a margin account with the clearinghouse.

The operation of margins

- A margin is cash or marketable securities deposited by an investor with his or her broker.
- The balance in the margin account is adjusted to reflect daily settlement.
- Margins minimise the possibility of a loss through a default on a contract.

Daily settlement

- An investor takes a long position in one December 2010 ASX200 futures contract on October 20
  - Contract size is AUD 115,900 (=4636*25)
  - Futures price is 4,636.0 index points
  - Margin requirement is AUD 3,250/contract
  - Maintenance margin is equal to the initial margin (AUD 3,150)

Daily settlement

- Operation of margins for a long position in the ASX 200
- Index Futures contract

Day	Futures price (index point)	Daily gain (loss) (AUD)	Cumulative gain (loss) (AUD)	Margin account balance (AUD)	Margin call (AUD)
	4636.0			3250	
20 October	4632.0	(100)	(100)	3150	
21 October	4630.0	(50)	(150)	3100	150
22 October	4649.0	475	325	3725	
25 October	4719.0	1750	2075	5475	
26 October	4689.0	(750)	1325	4725	
27 October	4643.0	(1150)	175	3575	

The over-the-counter market

- The over-the-counter (OTC) market is an important alternative to exchanges
- It is a telephone and computer-linked network of dealers who do not physically meet
- Trades are usually between financial institutions, corporate treasurers and fund managers

Forward contracts

- A forward contract is an OTC agreement to buy or sell an asset at a certain time in the future for a certain price.
- There is no daily settlement.
- At the end of the life of the contract one party buys the asset for the agreed price from the other party.

Forward contracts vs. futures contracts

Forward	Futures
Private contract between two parties	Traded on an exchange
Not standardized	Standardized
Usually one specified delivery date	Range of delivery dates
Settled at end of contract	Settled daily
Delivery or final settlement usual	Usually closed out prior to maturity
Some credit risk	Virtually no credit risk

https://www2.asx.com.au/markets/trade-our-derivatives-market/derivatives-market-prices/grain-derivatives

Regulation

- Regulation is designed to protect the public interest
- Regulators try to prevent questionable trading practices by individuals on the floor of the exchange or outside groups
- Australian Regulators
  - ASIC: Australian Securities and Investments Commission
  - ACCC: Australian Competition and Consumer Commission

Next Week

Hedging with Futures

- Short/Long Hedges
- Basis Risk
  - Cross Hedging
- Use of Stock Index Futures
- Hedge Ratio